Data Preprocessing

ifo Business Survey Data - Two Version Comparison

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Two versions prepared:

- **Version A**: Drop BES & PRS, use all years (1991-2025)
- **Version B**: Keep all questions, start from 1997

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1. Setup and Data Loading

Dimensions: 1088550 rows × 9 columns

Date range: 1991-01 to 2025-02

Industries: 177

Questions: 15

Data Structure

```
tibble [1,088,550 x 9] (S3: tbl_df/tbl/data.frame)
 $ date         : Date[1:1088550], format: "1991-01-01" "1991-01-01" ...
 $ industry_code: chr [1:1088550] "C0000000" "C0000000" "C0000000" "C0000000" ...
 $ level        : Named int [1:1088550] 0 0 0 0 0 0 0 0 0 0 ...
  ..- attr(*, "names")= chr [1:1088550] "C0000000" "C0000000" "C0000000" "C0000000" ...
 $ question     : chr [1:1088550] "KLD" "GUS" "GES" "LUS" ...
 $ value        : num [1:1088550] 8.325 19.735 -2.493 -0.116 -8.922 ...
 $ industry_name: Named chr [1:1088550] "Verarbeitendes Gewerbe" "Verarbeitendes Gewerbe" "Verarbeitende
  ..- attr(*, "names")= chr [1:1088550] "C0000000" "C0000000" "C0000000" "C0000000" ...
 $ question_name: Named chr [1:1088550] "Geschäftsklima" "Geschäftslage" "Geschäftserwartungen" "Kapazi
  ..- attr(*, "names")= chr [1:1088550] "KLD" "GUS" "GES" "LUS" ...
 $ question_type: Named chr [1:1088550] "Composite" "Current" "Expectation" "Current" ...
  ..- attr(*, "names")= chr [1:1088550] "KLD" "GUS" "GES" "LUS" ...
 $ variable     : chr [1:1088550] "C0000000_KLD" "C0000000_GUS" "C0000000_GES" "C0000000_LUS" ...
```

date	industry_code	industry_name	level	question	question_name	value
1991-01-01	C0000000	Verarbeitendes Gewerbe	0	KLD	Geschäftsklima	8.32469
1991-01-01	C0000000	Verarbeitendes Gewerbe	0	GUS	Geschäftslage	19.73534

1991-01-01	C0000000	Verarbeitendes Gewerbe	0	GES	Geschäftserwartungen	-2.49343
1991-01-01	C0000000	Verarbeitendes Gewerbe	0	LUS	Kapazitätsauslastung	-0.11596
1991-01-01	C0000000	Verarbeitendes Gewerbe	0	BUS	Bestellungsbestand	-8.92210
1991-01-01	C0000000	Verarbeitendes Gewerbe	0	XUS	Auslandsauftragsbestand	-38.48721
1991-01-01	C0000000	Verarbeitendes Gewerbe	0	AVS	Auftragseingang	-1.78836
1991-01-01	C0000000	Verarbeitendes Gewerbe	0	BVS	Nachfrage gegen Vormonat	-9.89593
1991-01-01	C0000000	Verarbeitendes Gewerbe	0	QVS	Inlandsnachfrage gegen Vormonat	4.78709
1991-01-01	C0000000	Verarbeitendes Gewerbe	0	PVS	Verkaufspreise	6.71275

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2. Add Combined Variable Names

##`industrycode_questioncode`

Examples: - `C1000000_KLD` = Business climate for Food industry - `C2800000_KLD` = Business climate for Machinery industry

Total unique variables: 2655

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3. Data Quality Check

Overall Statistics:

- Total observations: **1,088,550**
- Missing observations: **45,690**
- Overall missing rate: **4.20%**

Missing by Question

Code	Question Name	Observations	Missing	Missing %	First Data
PRS	Produktionserwartung	72570	14657	20.20%	1997-07
BES	Beschäftigungserwartung	72570	14482	19.96%	1997-07
KLD	Geschäftsklima	72570	1584	2.18%	1991-01
PVS	Verkaufspreise	72570	1389	1.91%	1991-01
XUS	Auslandsauftragsbestand	72570	1336	1.84%	1991-01
BVS	Nachfrage gegen Vormonat	72570	1231	1.70%	1991-01
PWS	Verkaufspreise Erwartung	72570	1229	1.69%	1991-01
BUS	Bestellungsbestand	72570	1227	1.69%	1991-01
QES	Inlandsnachfrage Erwartung	72570	1223	1.69%	1991-01
AVS	Auftragseingang	72570	1222	1.68%	1991-01
GES	Geschäftserwartungen	72570	1222	1.68%	1991-01
GUS	Geschäftslage	72570	1222	1.68%	1991-01
LUS	Kapazitätsauslastung	72570	1222	1.68%	1991-01
QVS	Inlandsnachfrage gegen Vormonat	72570	1222	1.68%	1991-01

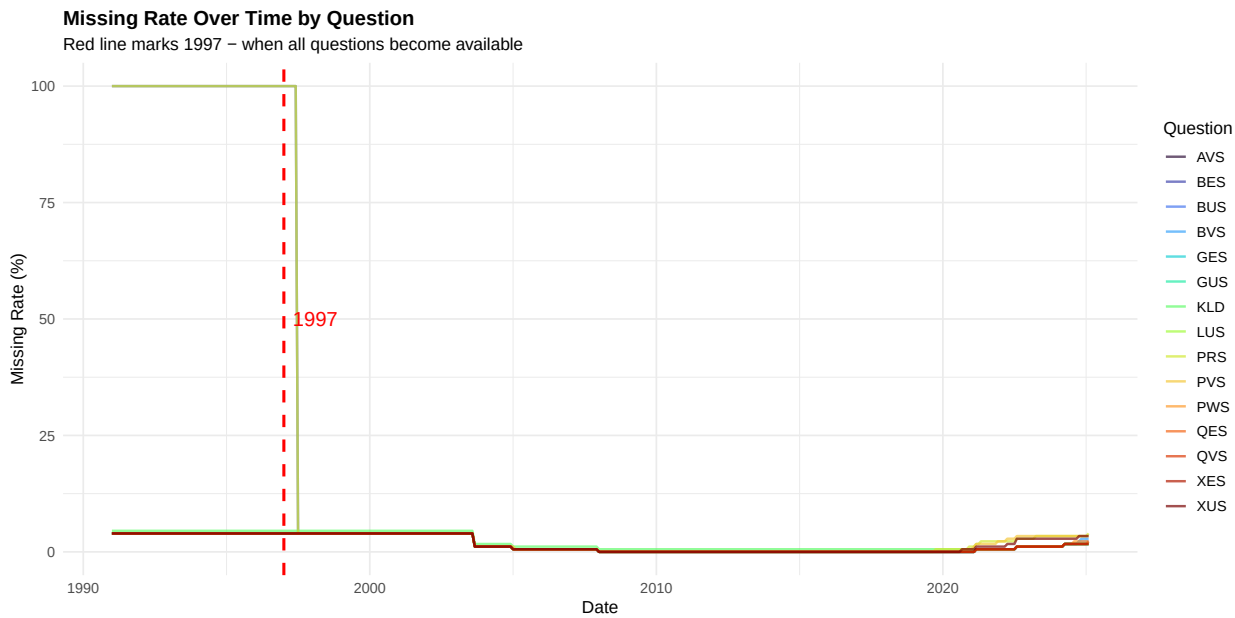
XES	Exporterwartungen	72570	1222	1.68%	1991-01

Key Findings:

Questions with >10% missing:

- PRS (Produktionserwartung): 20.20%, first data from 1997-07
- BES (Beschäftigungserwartung): 19.96%, first data from 1997-07

Visualize Missing Pattern



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4. Version Comparison

Defining two data preparation versions:

VERSION A: Drop BES & PRS, Use All Years

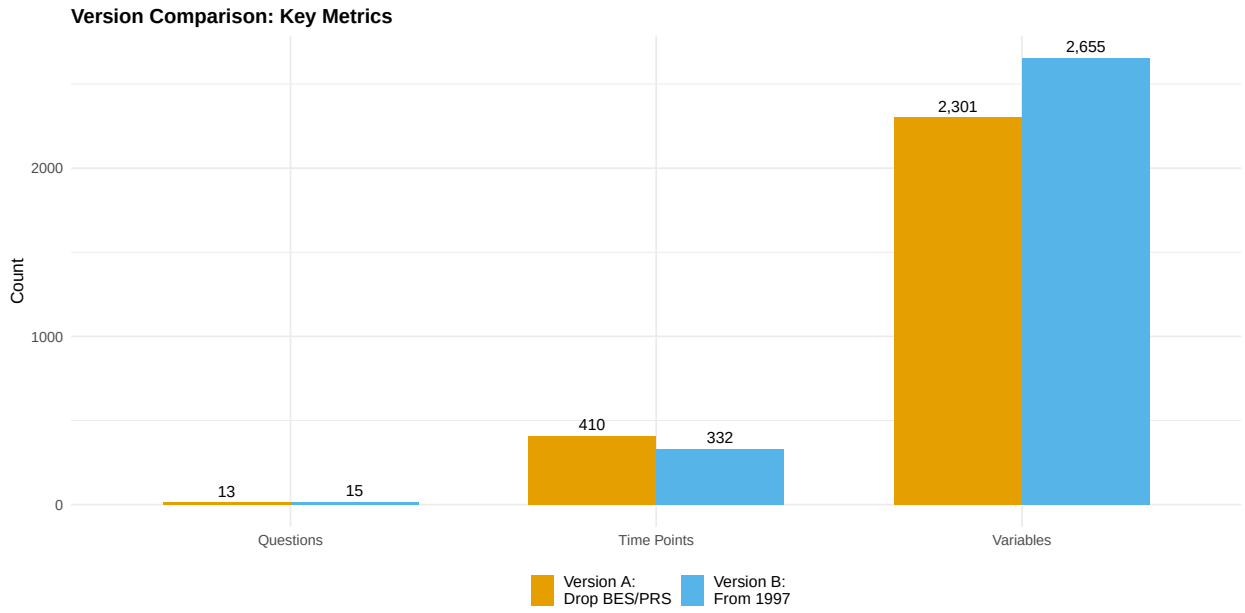
- Questions: 13 (exclude BES, PRS)
- Time period: 1991-01 onwards
- Rationale: Maximize time span

VERSION B: Keep All Questions, Start from 1997

- Questions: 15 (all questions)
- Time period: 1997-07onwards
- Rationale: Complete panel, include expectations

Metric	Version A	Version B (Preferred for DFM)
Time Period	1991-01 to 2025-02	1997-07 to 2025-02
Number of Months	410	332
Number of Questions	13	15
Number of Industries	177	177
Total Variables	2,301	2,655
Total Observations	943,410	881,460
Missing Rate	1.75%	1.24%

Visual Comparison



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5. Missing Value Patterns by Version

Version A: Missing by Question

Code	Name	Missing %	First Data
KLD	Geschäftsklima	2.18%	1991-01
PVS	Verkaufspreise	1.91%	1991-01
XUS	Auslandsauftragsbestand	1.84%	1991-01
BVS	Nachfrage gegen Vormonat	1.70%	1991-01
PWS	Verkaufspreise Erwartung	1.69%	1991-01
BUS	Bestellungsbestand	1.69%	1991-01
QES	Inlandsnachfrage Erwartung	1.69%	1991-01
AVS	Auftragseingang	1.68%	1991-01

GES	Geschäftserwartungen	1.68%	1991-01
GUS	Geschäftslage	1.68%	1991-01
LUS	Kapazitätsauslastung	1.68%	1991-01
QVS	Inlandsnachfrage gegen Vormonat	1.68%	1991-01
XES	Exporterwartungen	1.68%	1991-01

Version B: Missing by Question

Code	Name	Missing %	First Data
KLD	Geschäftsklima	1.63%	1997-07
PRS	Produktionserwartung	1.45%	1997-07
PVS	Verkaufspreise	1.43%	1997-07
XUS	Auslandsauftragsbestand	1.34%	1997-07
BVS	Nachfrage gegen Vormonat	1.17%	1997-07
PWS	Verkaufspreise Erwartung	1.16%	1997-07
BUS	Bestellungsbestand	1.16%	1997-07
QES	Inlandsnachfrage Erwartung	1.15%	1997-07
AVS	Auftragseingang	1.15%	1997-07
BES	Beschäftigungserwartung	1.15%	1997-07
GES	Geschäftserwartungen	1.15%	1997-07
GUS	Geschäftslage	1.15%	1997-07
LUS	Kapazitätsauslastung	1.15%	1997-07
QVS	Inlandsnachfrage gegen Vormonat	1.15%	1997-07
XES	Exporterwartungen	1.15%	1997-07

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6. Convert to Wide Format

Version A Wide Format

Wide format dimensions: 410 rows × 2302 columns

Version B Wide Format

Wide format dimensions: 332 rows × 2656 columns

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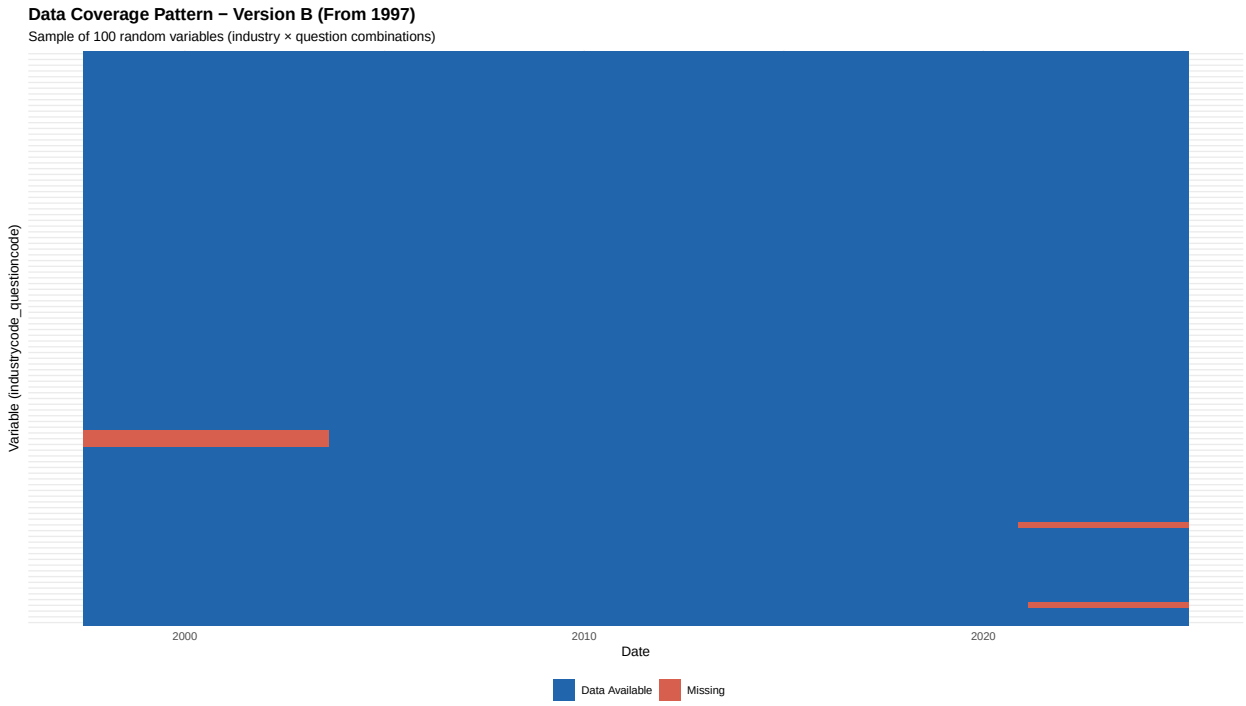
7. Data Coverage



Check for Issues

```
Version A - Completely missing series: 1

Version B - Completely missing series: 1

Version A - Series with >50% missing: 1

# A tibble: 1 x 2
  variable     missing_pct
  <chr>              <dbl>
1 C1020000_KLD         100



Version B - Series with >50% missing: 1

# A tibble: 1 x 2
  variable     missing_pct
  <chr>              <dbl>
1 C1020000_KLD         100
```

Business Climate Index for Fish processing industry is completely missing.

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8. Summary

Version A: Drop BES & PRS

- 410 months of data
- 13 questions
- 2,301 unique variables
- 1.75% missing rate

Version B: All Questions from 1997

- 332 months of data
- 15 questions (including BES & PRS)
- 2,655 unique variables
- 1.24% missing rate

Preference for DFM Analysis

Use Version B (All questions from 1997) because:

1. **Complete question set** (15 questions)
   - Includes BES (employment expectations)
   - Includes PRS (production expectations)
   - Both are forward-looking indicators
2. **Balanced panel structure**
   - All questions available for all time points
   - Better for DFM estimation
   - Easier to interpret factors
3. **Sufficient time span**
   - 332 months of monthly data
   - Exceeds literature standards (20-30 years)
   - Covers multiple business cycles
4. **Lower missing rate**
   - Version B: 1.24%
   - Better than Version A: 1.75%
5. **Cross-industry analysis**
   - Variable names (industrycode_questioncode) allow
   - Easy identification of industry correlations
   - Example: C1000000_KLD vs C2800000_KLD

Datasets

data_a_long.csv Dimensions: 943410 rows x 9 columns Key column: 'variable' (industrycode_questioncode)

Sample data:

date	variable	industry_name	question_name	value
1991-01-01	C0000000_KLD	Verarbeitendes Gewerbe	Geschäftsklima	8.32469
1991-01-01	C0000000_GUS	Verarbeitendes Gewerbe	Geschäftslage	19.73534
1991-01-01	C0000000_GES	Verarbeitendes Gewerbe	Geschäftserwartungen	-2.49343
1991-01-01	C0000000_LUS	Verarbeitendes Gewerbe	Kapazitätsauslastung	-0.11596
1991-01-01	C0000000_BUS	Verarbeitendes Gewerbe	Bestellungsbestand	-8.92210
1991-01-01	C0000000_XUS	Verarbeitendes Gewerbe	Auslandsauftragsbestand	-38.48721
1991-01-01	C0000000_AVS	Verarbeitendes Gewerbe	Auftragseingang	-1.78836
1991-01-01	C0000000_BVS	Verarbeitendes Gewerbe	Nachfrage gegen Vormonat	-9.89593
1991-01-01	C0000000_QVS	Verarbeitendes Gewerbe	Inlandsnachfrage gegen Vormonat	4.78709
1991-01-01	C0000000_PVS	Verarbeitendes Gewerbe	Verkaufspreise	6.71275

data_b_long.csv Dimensions: 881460 rows x 9 columns Key column: 'variable' (industrycode_questioncode)

Sample data:

date	variable	industry_name	question_name	value
1997-07-01	C0000000_KLD	Verarbeitendes Gewerbe	Geschäftsklima	7.72400
1997-07-01	C0000000_GUS	Verarbeitendes Gewerbe	Geschäftslage	3.94178
1997-07-01	C0000000_GES	Verarbeitendes Gewerbe	Geschäftserwartungen	11.57636
1997-07-01	C0000000_LUS	Verarbeitendes Gewerbe	Kapazitätsauslastung	8.72112
1997-07-01	C0000000_BUS	Verarbeitendes Gewerbe	Bestellungsbestand	-17.94559
1997-07-01	C0000000_XUS	Verarbeitendes Gewerbe	Auslandsauftragsbestand	-25.32635
1997-07-01	C0000000_AVS	Verarbeitendes Gewerbe	Auftragseingang	10.05003
1997-07-01	C0000000_BVS	Verarbeitendes Gewerbe	Nachfrage gegen Vormonat	4.96000
1997-07-01	C0000000_QVS	Verarbeitendes Gewerbe	Inlandsnachfrage gegen Vormonat	3.74152
1997-07-01	C0000000_PVS	Verarbeitendes Gewerbe	Verkaufspreise	-0.91541

data_a_wide.csv Dimensions: 410 rows x 2302 columns

Format: Each column is a variable (industrycode_questioncode)

Missing rate: 1.75%

data_b_wide.csv Dimensions: 332 rows x 2656 columns

Format: Each column is a variable (industrycode_questioncode)

Missing rate: 1.24%

Exported Files:

- `data_a_long.csv`: Version A long format (Drop BES/PRS)
- `data_a_wide.csv`: Version A wide format (Drop BES/PRS)
- `data_b_long.csv`: Version B long format (All questions from 1997)
- `data_b_wide.csv`: Version B wide format (All questions from 1997)